Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S) & LIIT | Paper -2(Adv-2025-P2-Model) | Date: 28-12-2025

Time: 02.00Pm to 05.00Pm | GTA-09 | Max. Marks: 180

KEY SHEET

MATHEMATICS

1	**A**	2	**C**	3	**C**	4	**D**	5	**BCD**	6	**ABC**
7	**BCD**	8	**AD**	9	**3**	10	**4040**	11	**4**	12	**6.67-6.73**
13	**3960**	14	**0**	15	**4.50**	16	**230**				

PHYSICS

17	**A**	18	**D**	19	**A**	20	**C**	21	**ACD**	22	**AD**
23	**CD**	24	**BC**	25	**6.40**	26	**8**	27	**8**	28	**1.9 - 2**
29	**2**	30	**3**	31	**1**	32	**8**				

CHEMISTRY

33	**C**	34	**C**	35	**A**	36	**C**	37	**BD**	38	**ACD**
39	**BCD**	40	**ACD**	41	**4**	42	**204**	43	**20**	44	**114**
45	**3**	46	**8**	47	**105**	48	**5**				

For More Join: @IITJEE_Advanced

SOLUTIONS

MATHEMATICS

1.

2. $N = 990$

3. $\tan \pi x \geq 1 \;\&\; x \in (-\infty,-3) \cup \left(-\frac{1}{2},1\right) \cup \left(\frac{3}{2},\infty\right)$

4. Let $u = \left|z^2\right|\left|z^2 + \frac{1}{z^2} + z + \frac{1}{z} - 2I\right| = \left|\left(z^2 + \bar{z}^2\right) + \left(z - \bar{z}\right) - 2i\right|$

$u = 2\left|2x^2 + x - 1 - i\right|$

$u^2 = 4((2x^2 + x - 1)^2 + 1)$

$\because |z| = 1 \qquad \therefore x^2 + y^2 = 1$

$-1 \leq x \leq 1$

Now $t = 2x^2 + x - 1 = 2\left(x^2 + \frac{1}{2}x - \frac{1}{2}\right)$

$= 2\left(\left(x + \frac{1}{4}\right)^2 - \frac{9}{16}\right) \qquad \frac{-9}{8} \leq t \leq 2 \qquad u_{\max}^2 = 20$

$u_{\max} = \sqrt{20}$

5. $I = \int \frac{4}{(3-2x)^2}\left(\frac{3+2x}{3-2x}\right)^{\frac{1}{3}} dx \qquad$ Putting $\frac{3+2x}{3-2x} = z^3$

We get $\frac{4}{(3-2x)^2}dx = z^2.dz \qquad \therefore I = \int z^2.z^{-1}dz = \frac{z^2}{2} + C = \frac{3}{6}\left(\frac{3+2x}{3-2x}\right)^{\frac{2}{3}} + C$

$\therefore \alpha = 6, \beta = 2, \gamma = 3$

6.

7. $0 <^2 \frac{\left(\tan^{-1}x - x\right)}{x^2} < 1$ as $\tan^{-1}x < \frac{x^2}{2} + x$. $\forall x < 0$ as $f(x) = \frac{x^2}{2} + x - \tan^{-1}x \;\&$

$f'(x) = x + 1 - \frac{1}{1+x^2} = 0$ at $x = 0 \;\&\; f''(0) > 0 \Rightarrow f(x) \geq f(0)$

$\Rightarrow f(x) = \begin{cases} 0; & x < 0 \\ \left(x^2 - 4\right)\left|x^2 - 2x\right|; & x \geq 0 \end{cases} \Rightarrow$ cont. $\forall x \in R$ but not diff. at $x = 0$.

8. "$x = 0$" is not in domain & $f(a) = -a$ as $\tan^{-1}t + \tan^{-1}\frac{1}{t} = \frac{\pi}{2} \forall t > 0$

$= -\frac{\pi}{2} \forall t < 0$

$-x = |x| + |x+1| - 1 \Rightarrow$ only o is 1 solution as included, & $f(4) = -4$.

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9. $x^3+x^2-px+q=(x-x_0)\left(x^2-qx+2\right)$

$I=-x_0-q \quad -P=2+qx_0$

$q=-2x_0 \quad \Rightarrow x_0=1, q=-2, P=0$

10. $2\alpha+\beta=\dfrac{-3^{2020}}{2^{2022a}} \quad 2\alpha'+\beta'=\dfrac{-3^{2020}}{2^{2022a}} \quad \alpha'+\beta'=\dfrac{-2.3^{2020}}{3.2^{2022a}}$

11. By the given condition we have $(a-r-1)^2=r^2$. Thus

$4a\le b^2=r^2-(a-r-1)^2=2r(a-1)-(a-1)^2$

That is $a^2-2(r-1)a+2r+1\le 0$

The above quadratic inequality of a has solutions, therefore the discriminant

$\left[2(r-1)\right]^2-4(2r+1)=4r(r-1)\ge 0,$

The solution is $r\ge 4$.

After checking , $(a,b)=\left(3,2\sqrt{3}\right)$ satisfies the given condition when $r\ge 4$. Therefore , the minimum value of r is 4.

12. Without loss of generally , let $\vec{e}=(1,0)$. Since $\vec{a}.\vec{e}=2$, we can put $\vec{a}=(2,s)$. Then for any real number t, we have

$4+s^2=|\vec{a}|^2\le 5|\vec{a}+t\vec{e}|=5\sqrt{(2+t)^2+s^2}$

This is equivalent to $4+s^2\le 5|s|$. So the solution is $|s|\in[1,4]$,namely

$s^2\in[1,16]$

Hence, we obtain $|\vec{a}|=\sqrt{4+s^2}\in\left[\sqrt{5},2\sqrt{5}\right]$.

13. It is easy to see that there are $\binom{4}{2}.\ P_2^4=72$ different ways to put two "a"s into the matrix such that each row and each column contains at most one "a". Similarly, there are also $\binom{4}{2}.\ P_2^4=72$ different ways for two "b"s to do the same thing. By the multiplicative principle we get 72^2 ways. Among them.

We exclude 72 cases in which two "b"s occupy the same cells as the two "a"s do, and exclude $\binom{16}{1}.\ P_2^9=16\times 72$ cases in which one "a"s shares the same cell with a "b".

Therefore there are $72^2-72-16\times 72=3960$ different ways that meet the requirement.

14. $(\sin x+\sqrt{3}\cos x)\sin x\cos y=2$

Equality exists if $\sin x+\sqrt{3}\cos x=\pm 2$ and

Corresponding $\sin x\cos y=\pm 1$

Which is impossible.

15. Let $\left|\overrightarrow{OA}\right|=2$

$\left|\overrightarrow{OB}\right|=4$

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$$\left|\overrightarrow{OC}\right| = 6$$

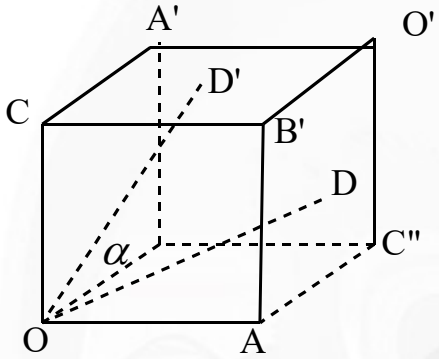
$$\overrightarrow{OD} = \overrightarrow{OA} + \frac{1}{2}\overrightarrow{AC'} + \frac{1}{2}\overrightarrow{AB'}$$

$$= 2\hat{i} + 2\hat{j} + 3\hat{k}$$

$$\overrightarrow{OD} = \overrightarrow{OC} + \frac{1}{2}\overrightarrow{CB'} + \frac{1}{2}\overrightarrow{B'O'}$$

$$= \hat{i} + 2\hat{j} + 6\hat{k}$$

$$\cos\alpha = \frac{\overrightarrow{OD}.\overrightarrow{OD'}}{\left|\overrightarrow{OD}\right|\left|\overrightarrow{OD'}\right|} = \frac{24}{\sqrt{17.41}}$$



16. Notice that we must have

$$\frac{f(f(z))-f(z)}{f(z)-z} = -\frac{f(f(z))-f(z)}{z-f(z)} \in i\mathbb{R}.$$

However $f(t)-t = t(t-20)$, so

$$\frac{f(f(z))-f(z)}{f(z)-z} = \frac{(z^2-19z)(z^2-19z-20)}{z(z-20)}$$

$$= \frac{z(z-19)(z-20)(z+1)}{z(z-20)}$$

$$= (z-19)(z+1)$$

$$= (z-9)^2 - 100.$$

Then, the real part of $(z-9)^2$ is 100. Since $\mathrm{Im}(z-9) = 11$, let $z-9 = a+11i$. Then,

$100 = \mathrm{Re}((a+11i)^2) = a^2 - 121 \Rightarrow a = \pm\sqrt{221}$

It follows that $z = 9+\sqrt{221}+11i$, and the requested sum is $9+221 = 230$

(solution by The Ultimate123)

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PHYSICS

17. Moment of inertia of combined system about y-axis

$$= \frac{4m\left(\frac{R}{2}\right)^2}{4} + \frac{mR^2}{4} + \frac{mR^2}{2} + mR^2 = 2mR^2$$

Conservation of angular momentum along y-axis $mv_0 \frac{R}{2} = 2mR^2\omega \quad \omega = \frac{v_0}{4R}$

18. $eV = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{\frac{2eV}{m}}$ *and* $T = \frac{2\pi m}{eB}$ So $S = vT = \frac{2\pi m}{eB}\sqrt{\frac{2eV}{m}}$ $\left[S \Rightarrow 2\sqrt{\frac{20mV}{eB^2}}\right]$

19. Charge particle will move in a helical path

20. Conceptual

21. As the ammeters are ideal, so $R_{eq} = \frac{R_1R_2R_3}{R_1R_2 + R_2R_3 + R_3R_1}$

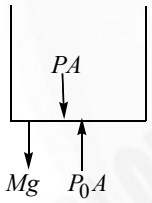
The readings of two ammeters are different so resistors R_1 *and* R_2 are different.

If R_1 *and* R_3 are interchanged, the reading of ammeters will also change.

If $R_2 = R_1$ (If R_2 *and* R_1 switched the reading of the ammeter do not vary) similarly if $R_2 = R_3$ (R_2 *and* R_3 switched the reading of ammeters do not change)

If $R_2 = R_3$, $I_2 = I_3 = 0.10, I_1 = 0.20$, and $I = 0.40\,amp$.

22.

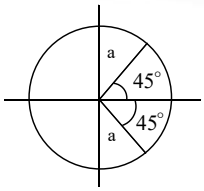


$Mg + PA = P_0A \quad M = \frac{P_0A}{g} - \frac{PA}{g} \quad P = \frac{P_i}{2} = \frac{P_0 + \frac{mg}{A}}{2} \Rightarrow M = \frac{P_0A}{2g} - \frac{m}{2}$

23. $y = a_0\sin(kx)\sin(\omega t + \phi)$ $\frac{kL}{3} = n\pi$ and $\frac{kL}{2} = \left(n + \frac{1}{2}\right)\pi$.

$\Rightarrow n = 1, k = \frac{3\pi}{L}$ Also $a' = \left|a_0 \sin\left(\frac{3\pi}{L} \times \frac{5L}{6}\right)\right| = a_0$

24.



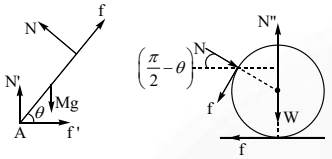
$\frac{-GMm}{2R} = \frac{-GMm}{R} + \frac{1}{2}mv_0^2 \quad \Rightarrow v_0 = \sqrt{\frac{GM}{R}}$

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While in the tunnel: $\frac{d^2x}{dt^2} = -\left(\frac{GM}{R^3}\right)x$

$$v_0^2 = \frac{GM}{R} = \omega^2(a^2 - R^2) \Rightarrow a = \sqrt{2}R \quad \Rightarrow T_{A\to B} = \frac{1}{4}\times 2\pi\sqrt{\frac{R^3}{GM}}$$

25.



$\tau_A = 0 \qquad N = \frac{Mg}{2}\cos\theta$ For disc, $N\sin\theta = f + f\cos\theta$

$M = \lambda L$ and $L = \frac{R}{\tan\frac{\theta}{2}} \qquad f = \frac{1}{2}\lambda g R\cos\theta \qquad = \frac{1}{2}\times 8\times 10\times\frac{2}{10}\times\frac{4}{5} = \frac{32}{5} = 6.40\ N$

26. $I\alpha\frac{1}{I^2}$ where I: distance from point source $\Rightarrow \frac{I_A}{I_B} = \frac{I_B^2}{I_A^2} = 2 \Rightarrow I_B = \sqrt{2I_A}$.....(1)

Also , due to polaroids : $I_B = \frac{I_B}{2}\cos^2 45^0 = \frac{I_B}{4} \Rightarrow I_B = \frac{I_B}{4}$.....(2)

$\Rightarrow$ *Ratio becomes* 4 *times* $\Rightarrow r_{new} = 8$

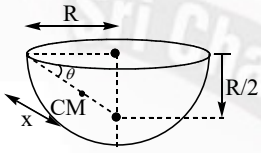
27. $F = 2m\eta A\left[(v_0+v)^2 - (v_0-v)^2\right] = 2\left(10^{-26}\right)\left(10^{25}\right)(1)\left[(5+2)^2 - (5-2)^2\right] = 0.2\times 40 = 8N$

28. Conceptual.

29. Amplitude of block $= \frac{mg}{k} = 1mm \qquad \frac{1}{v} + \left(\frac{1}{-30}\right) = \frac{-1}{20} \quad \Rightarrow v = -60\ cm$

Magnification, $m = -\frac{v}{u} = -2$ This; amplitude of image = 2mm.

30. $V_y = \omega x\cos\theta = \frac{\omega R}{2}$. Using conservation of angular momentum about centre of mass.



$$mu\frac{R}{4} = \left\{\frac{2}{3}mR^2 - \frac{mR^2}{4} + \frac{5mR^2}{16}\right\}\omega + \frac{5mR^2}{16}\omega \qquad = \frac{25}{24}mR^2\omega \Rightarrow \omega = \frac{6u}{25R}\ V_y = \frac{3u}{25}$$

31. Required mass $= \frac{10^9}{Q-value\times 0.5}\times 2\times m\left(\frac{2}{1}D\right)\times 24\times 3600 = 1.3kg$

32. $j = \frac{1}{\rho}E = \frac{\sigma}{2\rho\varepsilon} \qquad I = jA = \frac{q}{2\rho\varepsilon} \Rightarrow \frac{dp}{dt} = -\frac{q}{2\rho\varepsilon} \quad \Rightarrow q = Qe^{-\left(\frac{1}{2\rho\varepsilon}\right)}$ Or $q = \frac{Q}{8}$ at $r = 6\rho\varepsilon\ \ell n(2)$

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CHEMISTRY

33. NGP takes place, product formed through achiral intermediate

34. After mixing

$C_1 = 0.1M$

$C_2 = 0.1M$

$\left[H^+\right] = \sqrt{Ka_1C_1 + Ka_1C_2}$

$= \sqrt{3\times10^{-5}\times0.1 + 10^{-5}\times0.1} \qquad = \sqrt{4\times10^{-6}}$

$= 2\times10^{-3} \qquad p^H = 3 - \log 2$

35. (i) Ce^{3+} to Lu^{3+}: Ionic radius decreases

(ii) Eu: Metallic radius = 199 Pm

(iii) The hydroxides are definite compounds, not just hydrated oxides.

(iv) $E^0_{Ce^{4+}/Ce^{3+}} = +1.74V$.

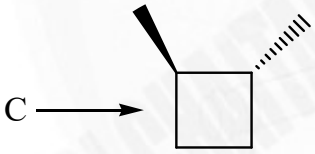
36. (i) $BaCrO_4$:yellow

(ii) $Fe^{+2} \xrightarrow{CN^-} Fe(CN)_2\downarrow \xrightarrow{CN^-} \left[Fe(CN)_4\right]^{2-}(yellow)$

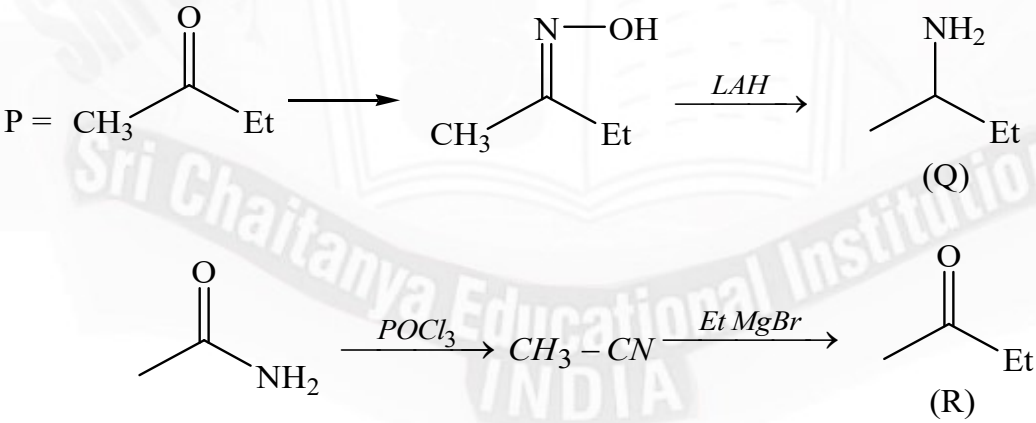
(iii) $Ag^+ \xrightarrow{CrO_4^{2-}} Ag_2CrO_4\downarrow(red)$

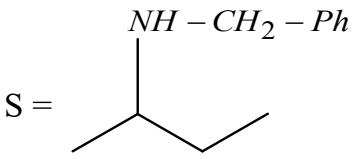
(iv) $NH_4^+ \xrightarrow{H_2[PtCl_6]} (NH_4)_2\left[PtCl_6\right]$: yellow

37.



38.





39. For the given reaction

ΔH is positive

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$$\ln K_p = \frac{\Delta S^0}{R} - \frac{\Delta H^0}{R}.\frac{1}{T}$$

$$\text{Slope} = \ln K_p = -\frac{\Delta H^0}{R} = -ve$$

$$\text{Intercept} = \frac{\Delta S^0}{R} = +ve$$

K_p Increases with the increase of temperature

An endothermic reaction which is spontaneous at high temperature becomes nonspontaneous at low temperature

40. N_2O_3, N_2O_4 :planar molecules

41.

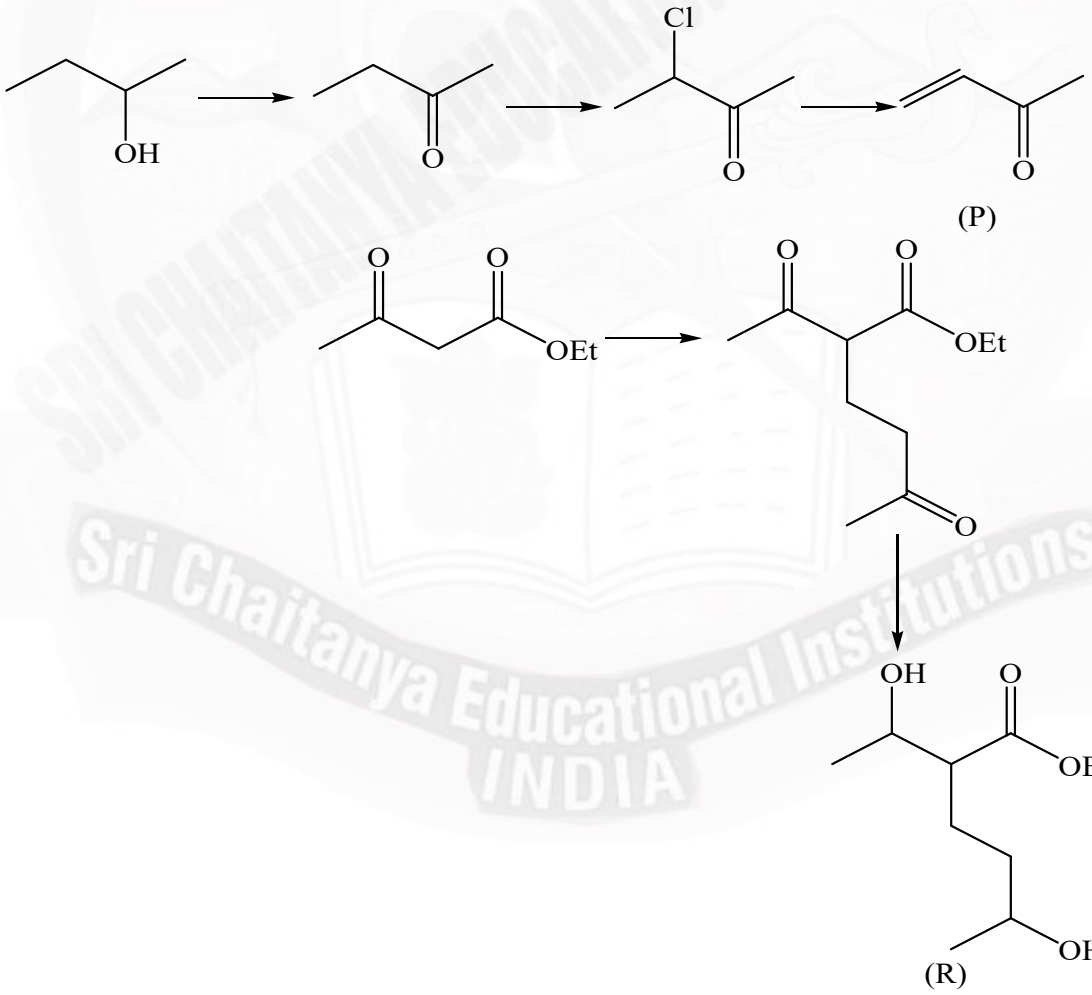
Ala-Leu-Phe-Gly

Ala-Leu-Gly-Phe

Ala-Phe-Leu-Gly

Ala-Phe-Gly-Leu

42.



43. Let m eq of $CaCO_3$ in 10ml solution $=x$

M eq of $Ca(HCO_3)_2$ in 10ml solution=y

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In presence of Hph:

$\frac{x}{2} = 10 \times 0.05 \times 2$

$x = 2$

In presence of $MeOH$

$x + y = 32.35 \times 0.05 \times 2$

$y = 1.235$

In 100ml

M eq of $CaCO_3 = 20$

$\therefore wt = 1 gram$

M eq of $Ca(HCO_3)_2 = 12.35$

$\therefore wt = 1 gram$

Wt of $NaCl = 0.5 gram$

$\%\ NaCl = \frac{0.5}{2.5} \times 100 = 20\%$

44. Volume displaced by piston

= Volume change of $Cl_2(g)$

$= 500 cm^2 \times 48 cm$

$= 24 l$

Moles of Cl_2 consumed in the reaction

$n = \frac{PV}{RT} = \frac{1 \times 24}{0.08 \times 300}$

n = 1

ΔH^o for the above process

$= \frac{n \times \Delta H^o}{3} = \frac{1 \times 342}{3} = 114\ kCal$

45. Given $E_n - E_1 = 108.9 eV$

$\left(-13.6 \times \frac{Z^2}{n^2}\right) - \left(-13.6 \times \frac{Z^2}{1^2}\right) = 108.9 eV$

$13.6\ Z^2 \left[1 - \frac{1}{n^2}\right] = 108.9\ eV$ ....(i)

Given $\lambda_n - \lambda_1 = 2\pi r_1 \times 2$

$\left(\frac{2\pi r_n}{n} - \frac{2\pi r_1}{1}\right) = 2 \times 2\pi r_1$ $(by\ 2\pi r_n = n\lambda)$

$\frac{2\pi r_1 n^2}{n} - 2\pi r_1 = 4\pi r_1$

$\therefore\ n - 1 = 2$

$n = 3$

Putting n back in eq. (i)

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$\boxed{Z=3}$

46. $T_b = \dfrac{a}{bR} = \dfrac{4.105}{0.1\times 0.0521}$

$= 500K$

Real gas behaves like an ideal gas

$d = \dfrac{PM}{RT}$

$D = 8\ gram/litre$

47. $X = MnCl_2,\ Y = H_2S,\ Q = Mn(OH)_2,\ Z = MnO_2.H_2O / MnO(OH)_2$

48. $X = \left[Ni(H_2O)_6\right] Cl_2$

$P = \left[Ni(H_2O)_4 en\right]Cl_2 : \left[Ni(H_2O)_4(en)\right]^{2+}$

$Q = \left[Ni(H_2O)_2(en)_2\right]Cl_2 : \left[Ni(H_2O)_2(en)_2\right]^{2+}$: 3 isomers

$R = \left[Ni(en)_3\right]Cl_2 : \left[Ni(en)_3\right]^{2+}$: 2 isomers

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